Tutorial VII - Parcel Delivery Routing

Applied Optimization with Julia



Introduction

Welcome to “FastDelivery” - a local logistics service in Hamburg! We’re working with a central warehouse that operates a decentralized distribution network: they deliver packages directly to smaller distribution centers in the region, ensuring efficient last-mile delivery across the area.

The central warehouse operates a fleet of delivery vehicles that transport packages between their main hub and 14 partner distribution centers scattered across the region. The challenge? Designing the most efficient routing system to keep packages flowing smoothly while keeping operational costs in check.

In this case study, we’ll help FastDelivery optimize their delivery routes using the Capacitated Vehicle Routing Problem (CVRP) approach. We’ll consider several real-world aspects like vehicle capacity (measured in parcels), driving distances, and operational costs to find the best possible delivery strategy. The driving time restriction from the lecture is not part of this tutorial.

In this tutorial, you will:

1. Implement the basic CVRP model with degree and depot constraints
2. Diagnose subtours visually in a plotted solution
3. Add MTZ constraints to eliminate subtours and respect capacities
4. Translate a solution into monthly operational costs
5. Optimize the fleet size based on costs

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1. Implementing the basic CVRP

Implement the CVRP from the lecture without the constraints preventing subtours and restricting the capacities. The demand per distribution center and the coordinates of the locations are provided as CSV files.

The number of vehicles is 6 and each vehicle can transport up to 50 parcels to the distribution centers. Note that the capacity of each vehicle is identical.

Loading the data

First, we need to load the required packages and data files. Make sure you have the following files in the `data` directory that is located in the same directory as this tutorial file:

- `routing-demand.csv`: Contains demand per distribution center

- `coord_x_y.csv`: Contains location coordinates

```
using CSV
using DataFrames
using JuMP
using HiGHS
using Plots

## Prepare the model data
file_directory = "$(@__DIR__)/data"
demand = CSV.read("$(file_directory)/routing-demand.csv", DataFrame)
coord = CSV.read("$(file_directory)/coord_x_y.csv", DataFrame)
```

```
15×3 DataFrame
 Row │ location   x         y
     │ DataStri…  Float64   Float64
─────┼──────────────────────────────
   1 │ central    155.24    151.63
   2 │ dc_01      128.513    67.4002
   3 │ dc_02       47.6949  194.738
   4 │ dc_03       93.6483  181.213
   5 │ dc_04       59.737   166.434
   6 │ dc_05      197.681   102.756
   7 │ dc_06      152.263   140.986
   8 │ dc_07      183.943   182.649
   9 │ dc_08      113.792    33.3626
  10 │ dc_09      172.555    82.9658
  11 │ dc_10       77.4372   27.0503
  12 │ dc_11      132.689   152.717
  13 │ dc_12       54.2646    0.0399352
  14 │ dc_13       54.5474   31.4915
  15 │ dc_14       66.5128  100.22
```

> **Tip**
> Make sure that you have downloaded the data in the `data` folder that is located in the same directory as this tutorial file.

The coordinates are given in kilometers, so the Euclidean distance between two locations is the driving distance in kilometers.

Define the parameters

First, define the number of vehicles and the capacity as parameters. Name them `vehicles` and `capacity`.

```
# YOUR CODE BELOW
```

```
# Test your code
@assert vehicles == 30/5 "Number of vehicles is incorrect"
@assert capacity == 100/2 "Capacity is incorrect"
println("Variables defined, great job!")
```

Now, we need to compute the distance matrix. You can do so by using the Euclidean distance between the coordinates of the locations. Compute the distance matrix for all locations and store it in a dictionary called `distance`. Access each location by the tuple `(i,j)` where `i` is the origin and `j` is the destination. **Hint: You need the distance from each location to all other locations! Also include the pairs `(i,i)` with a distance of 0, so the dictionary has an entry for every combination.**

```
# YOUR CODE BELOW
```

```
# Test your code
@assert length(distance) == nrow(coord) * nrow(coord) "Distance dictionary
should have n² entries where n is the number of locations"
@assert all(v >= 0 for v in values(distance)) "All distances should be non-
negative"
@assert all(isapprox(distance[(i,j)], distance[(j,i)]) for i in coord.location
for j in coord.location if i != j) "Distances should be symmetric"
println("Distance dictionary test passed successfully, great job!")
```

Define the model

Next, we create the model instance. As this is a linear problem, we can use the `HiGHS` solver.

```
# Create the model instance
cvrp_model = Model(HiGHS.Optimizer)
set_attribute(cvrp_model, "presolve", "on")
set_attribute(cvrp_model, "time_limit", 60.0)
set_attribute(cvrp_model, "mip_rel_gap", 0.0)
```

Define the variables

Now you can start by defining the model variables. The variable we need first is the binary decision variable `X` which indicates whether an arc from node `i` to node `j` exists. To work with the decision variable, you will need to work with a vector of locations.

```
# Vector of locations
println(demand.location)
```

```
DataStrings.DataString["central", "dc_01", "dc_02", "dc_03", "dc_04", "dc_05",
"dc_06", "dc_07", "dc_08", "dc_09", "dc_10", "dc_11", "dc_12", "dc_13",
"dc_14"]
```

> **Tip**
> You could use the `demand.location` vector to define the elements of the decision variable `X`.

```
# YOUR CODE BELOW
```

In the lecture, `X` was only defined for real arcs with `i != j`. Our matrix variable also contains the "self-loops" `X[i,i]` and nothing in the later constraints forbids the solver from setting them to 1 at zero cost. To rule this out, the following provided code fixes all diagonal entries to zero:

```
# Self-loops are not part of any tour, thus we fix them to zero
for i in demand.location
    fix(X[i,i], 0; force=true)
end
```

```
# Test your code
@assert size(X) == (15,15) "Have you defined the decision variable X?"
@assert all(is_binary(x) for x in X) "The decision variable X should be binary"
@assert all(is_fixed(X[i,i]) for i in demand.location) "The self-loops X[i,i]
should be fixed to zero - run the provided cell above!"
println("Decision variable X test passed successfully, great job!")
```

Define the objective function

We want to minimize the total distance of the routes based on the distance dictionary `distance` that you computed earlier. Create the objective function.

```
# YOUR CODE BELOW
```

```
# Test your code
obj = objective_function(cvrp_model)
@assert typeof(obj) <: GenericAffExpr "Objective should be a linear expression"
@assert length(obj.terms) > 0 "Objective should not be empty"
println("Objective function defined successfully, great job!")
println("Note that it is not tested whether the objective function is
correct!")
```

Define the constraints

Create the two sets of constraints that ensure that each location is visited exactly once. **Note that the central location is indexed as `"central"`!** We don't need to consider the central location for the constraints and we have to ensure during summing that we don't include the pairs where `i == j`.

```
# YOUR CODE BELOW
```

```
# Test your code
const_refs = all_constraints(cvrp_model,
include_variable_in_set_constraints=false)
num_constraints_in_model = length(const_refs)
expected_constraints = 2 * (nrow(demand) - 1)  # indegree + outdegree for all
nodes except central
@assert num_constraints_in_model == expected_constraints "Number of constraints
is incorrect as it is $(num_constraints_in_model) instead of
$(expected_constraints)."
println("Constraints for the inflow and outflow defined, great job!")
```

Next, we need to restrict the ingoing and outgoing flows to the `central` location. Create the two constraints that ensure that the number of vehicles arriving and leaving the central location is equal to the number of vehicles.

```
# YOUR CODE BELOW
```

```
# Test your code
const_refs = all_constraints(cvrp_model,
include_variable_in_set_constraints=false)
num_constraints_in_model = length(const_refs)
expected_constraints = 2 * (nrow(demand) - 1) + 2  # indegree + outdegree for
all nodes except central + 2 for inflow and outflow
@assert num_constraints_in_model == expected_constraints "Number of constraints
is incorrect as it is $(num_constraints_in_model) instead of
$(expected_constraints)."
println("Constraints for the inflow and outflow defined, great job!")
```

Solve the model

Now, we can solve the model. This should work very fast, as the model is very simple without any subtour elimination constraints. Solve the model as usual.

```
# YOUR CODE BELOW
```

```
# This defines a function that prints the status of the model
function print_model_status(model)
    begin
        if termination_status(model) == OPTIMAL
            println("Great, the solution is optimal.")
            println("The relative gap is $(relative_gap(model))")
            println("The solve time (in seconds) is $(solve_time(model))")
        elseif termination_status(model) == TIME_LIMIT && has_values(model)
            println("Solution is suboptimal due to a time limit, but a primal
solution is available")
        else
            error("The model was not solved correctly.")
        end
        println("The objective value is ", objective_value(model))
    end
end

# Test your code
@assert termination_status(cvrp_model) == OPTIMAL "The model should be optimal.
Have you made any mistakes in the model formulation?"
print_model_status(cvrp_model)
println("Model solved successfully, great job!")
```

Plot the results

The following part extracts the results to plot the tours and requires no changes on your part. It defines some functions that are used to plot the results.

```
function plot_routes(X, coord)

    Routes = [value(X[i,j]) >= 0.5 ? true : false for i in coord.location, j in
coord.location]

    fig = plot(
        size=(800, 600),
        background_color=:white,
        framestyle=:box,
        grid=false,
        dpi=300
    )

    connections = findall(Routes)

    for conn in connections
        xx = [coord[conn[1],:x]; coord[conn[2],:x]]
        yy = [coord[conn[1],:y]; coord[conn[2],:y]]
        plot!(
```

```
            fig, xx, yy,
            color=:crimson,
            label="",
            linewidth=2,
            linestyle=:solid,
            alpha=0.7
        )
    end

    scatter!(
        fig, coord[1:1,:x], coord[1:1,:y],
        marker=:star5,
        color=:black,
        markersize=15,
        label="Warehouse",
        markerstrokewidth=1
    )
    scatter!(
        fig, coord[2:end,:x], coord[2:end,:y],
        marker=:circle,
        color=:steelblue,
        markersize=8,
        label="Distribution Centers",
        markerstrokewidth=1
    )

    plot!(
        fig,
        title="Delivery Routes Optimization",
        xlabel="Distance (km)",
        ylabel="Distance (km)",
        legend=:bottomright,
        legendfontsize=8,
        foreground_color_legend=nothing,
        background_color_legend=:transparent,
        margin=5Plots.mm,
        xlims=(minimum(coord.x) - 5, maximum(coord.x) + 5),
        ylims=(minimum(coord.y) - 5, maximum(coord.y) + 5)
    )

    return fig
end

# Plot the routes
display(plot_routes(X,coord))
```

2. Subtour elimination

As you see in the plot, the solution contains subtours. Thus, we need to add some constraints from the Miller-Tucker-Zemlin (MTZ) formulation to prevent subtours. First, we need to define the variable `U` which is the current capacity usage at node `i`.

> **Tip**
> It might be useful to convert the `demand` DataFrame into a dictionary, so we can access the demand for each location directly by the location name for the variable definition and the constraints.

Create the variable `U` for all locations **except** `"central"`, as in the lecture. Define its bounds directly in the variable declaration: the capacity usage at each location has to be between the demand and the capacity.

```
# YOUR CODE BELOW
```

```
# Test your code
customers = [i for i in demand.location if i != "central"]
@assert all(has_lower_bound(U[i]) for i in customers) "The variable U should
have a lower bound for each distribution center"
@assert all(has_upper_bound(U[i]) for i in customers) "The variable U should
have an upper bound for each distribution center"
println("Variable U defined successfully, great job!")
```

Next, we need to add the set of constraints to prevent subtours.

```
# YOUR CODE BELOW
```

```
# Test your code
println("Constraints for the subtour elimination defined!")
println("It is not tested whether the constraints are correct, so please check
the resulting routes in the visualization.")
```

Now, solve the model again.

```
# YOUR CODE BELOW
```

The following code prints the model status and visualizes the routes. If your implementation is correct, the routes should be free of subtours and the model should have reached optimality or found a feasible solution before hitting the time limit.

```
print_model_status(cvrp_model)
display(plot_routes(X,coord))
```

The following code plots the capacity usage per vehicle. Each bar shows the value of `U` at the last stop before the vehicle returns to the warehouse.

> **Note**
> The bars are an **upper estimate** of the parcels actually carried: the MTZ constraints only force `U` to be *at least* as large as the accumulated demand, nothing pushes it down to the exact load. Can you see why from the constraints?

```
# Plot the capacity usage
function plot_capacity_usage(X, U, coord)
    # Find which locations are visited last by each vehicle before central
    vehicle_routes = []
    for i in coord.location
        if i != "central" && value(X[i, "central"]) > 0.5
            push!(vehicle_routes, (i, value(U[i])))
        end
    end

    # Sort by capacity usage for better visualization
    sort!(vehicle_routes, by = x -> x[2])

    # Create bar plot
    fig = bar(
        [u[2] for u in vehicle_routes],
        xticks=(1:length(vehicle_routes), [u[1] for u in vehicle_routes]),
        title="Capacity Usage per Vehicle",
        xlabel="Last stop before returning to the warehouse",
        ylabel="Capacity Usage (parcels)",
        label="",
        color=:steelblue,
        ylims=(0, capacity),
        size=(800, 400),
        dpi=300,
        margin=5Plots.mm,
    )

    # Add capacity limit line
    hline!([capacity], color=:red, linestyle=:dash, label="Capacity Limit")

    return fig
end

# Create and display the plot
display(plot_capacity_usage(X, U, coord))
```

---

3. Computing the costs

FastDelivery wants to know the costs of the optimal solution. Let's calculate the monthly operational costs based on the following parameters:

- Cost per kilometer: 0.6 EUR (includes fuel, repairs, driver costs)
- Service frequency: Monday to Friday (5 days per week)
- Time period: 4 weeks
- Vehicle lease cost: 450 EUR per vehicle per 4 weeks

Calculate the total monthly costs and store them in `monthly_costs`.

```
# YOUR CODE BELOW
```

```
# Test your code
@assert isapprox(monthly_costs, 14168, atol=10) "The costs are not as expected.
But don't worry, you can still get the bonus points!"
println("Costs calculated as $(monthly_costs) EUR successfully, great job!")
```

---

4. Optimal Solution?

Although the costs can now be computed, the tour plan has not changed so far. FastDelivery asks you whether you have an idea of how to lower the costs associated with the parcel distribution. Take a look at your current plotted solution and try to come up with an idea. Discuss the potential shortcoming and assumptions of the current model in a few sentences. Propose at least one idea of how to lower the costs in the future.

You can do this in a comment in the cell below.

```
#=




=#
```

---

5. Number of Vehicles

Together, you realize one potential area of improvement involves the number of vehicles: it might potentially not be necessary to have 6 vehicles which deliver the parcels to the distribution centers. You are thus tasked to find the optimal number of vehicles and the best tour plan to minimize the costs **(not the distance!)**.

Two questions worth answering before you start modelling:

1. **How much extra daily distance is one vehicle "worth"?** One leased vehicle costs 450 EUR per 4 weeks, while each kilometer of daily route length costs $0.6 \times 5 \times 4 = 12$ EUR per 4 weeks. Dropping a vehicle thus pays off unless the daily tours grow by more than $450/12 = 37.5$ km.

2. **What is the minimum feasible fleet size?** The total demand is 86 parcels and each vehicle can carry at most 50.

> **Tip**
> There are multiple ways to tackle this task, but I recommend first thinking about how to model the fixed costs for the vehicles and how to add the costs for the driving distance. If you need to adjust the model, feel free to do so. The easiest way then would be the definition of a new model (e.g. `cvrp_model_flex`) and then to copy and adjust all relevant constraints and variables.

Implement these changes in your model and compute the optimal number of vehicles and the best tour plan. In case the computation takes too long, no worries! If you define the model with a time limit of 60 seconds, it should not take too long.

```
# YOUR CODE BELOW
```

The code below again visualizes the results. Note that you might need to replace the `X` and `U` with your new model's variable names, if you have renamed them.

```
# Visualize the results
print_model_status(cvrp_model_flex)
display(plot_routes(X,coord))
```

Based on your results, what are the expected cost savings compared to the monthly costs from Section 3, and how many vehicles does FastDelivery need?

Please answer this question in the cell below.

```
#=



=#
```

Solutions

You will likely find solutions to most exercises online. However, I strongly encourage you to work on these exercises independently without searching explicitly for the exact answers to the exercises. Understanding someone else's solution is very different from developing your own. Use the lecture notes and try to solve the exercises on your own. This approach will significantly enhance your learning and problem-solving skills.

Remember, the goal is not just to complete the exercises, but to understand the concepts and improve your programming abilities. If you encounter difficulties, review the lecture materials, experiment with different approaches, and don't hesitate to ask for clarification during class discussions.

Bibliography